



Determination of the Binary Diffusion Coefficients and Interaction Viscosities of the Systems Carbon Dioxide–Nitrogen and Ethane–Methane in the Dilute Gas Phase from Accurate Experimental Viscosity Data Using the Kinetic Theory of Gases

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Received: 15 June 2023 / Accepted: 28 June 2023 / Published online: 29 July 2023
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Abstract

The results of viscosity measurements at moderate densities on the two gaseous mixtures carbon dioxide–nitrogen and ethane–methane including the pure gases between 253.15 K and 473.15 K, originally performed by Humberg *et al.* at Ruhr University Bochum, Germany, using a rotating-cylinder viscometer between 0.1 MPa and 2.0 MPa, were employed to determine the interaction viscosity, $\eta_{12}^{(0)}$, and the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, each in the limit of zero density. The isothermal viscosity data were evaluated by those authors with density series restricted to the second order at most to derive the zero-density viscosities and initial density viscosity coefficients, $\eta_{\text{mix}}^{(0)}$ and $\eta_{\text{mix}}^{(1)}$, for the mixtures, as well as, $\eta_i^{(0)}$ and $\eta_i^{(1)}$ ($i = 1, 2$), respectively, for the pure gases. Humberg *et al.* have already compared their $\eta_{\text{mix}}^{(0)}$ and $\eta_i^{(0)}$ data for carbon dioxide–nitrogen and ethane–methane with corresponding viscosity values theoretically computed for the nonspherical potentials of the intermolecular interaction. Now we employed $\eta_{\text{mix}}^{(0)}$ and $\eta_{\text{mix}}^{(1)}$ as well as $\eta_i^{(0)}$ and $\eta_i^{(1)}$ in two procedures to derive $\eta_{12}^{(0)}$ values. For this, we needed A_{12}^* values (ratio between effective cross-sections of viscosity and diffusion). But the second procedure applying the initial density viscosity coefficients $\eta_{\text{mix}}^{(1)}$ and $\eta_i^{(1)}$ failed to yield reasonable $\eta_{12}^{(0)}$ values. The first procedure should provide the best results when it is possible to use A_{12}^* values computed for the nonspherical potential. The effect is comparatively small if $\eta_{12}^{(0)}$ is determined. But if $(\rho D_{12})^{(0)}$ is calculated from $\eta_{12}^{(0)}$ using A_{12}^* values for the nonspherical potential, the impact is several percent. Moreover, the experimentally based $\eta_{12}^{(0)}$ and $(\rho D_{12})^{(0)}$ data agree with theoretically calculated values for the nonspherical potentials.

Keywords Gaseous mixtures · Interaction viscosity · Low-density viscosity values · Product molar density and diffusion coefficient · Viscosity · Zero-density limit

Special Issue in Honor of Professor Roland Span's 60th Birthday.

Extended author information available on the last page of the article

1 Introduction

The group of Professor Span at Ruhr University Bochum, Germany, has reported about viscosity measurements on gaseous mixtures including the respective pure gases. These measurements concerning the binary systems carbon dioxide–nitrogen [1] and ethane–methane [2] were performed up to 2 MPa in the temperature range from 253.15 K to 473.15 K using a rotating-cylinder viscometer and determining the decay rate of the cylinder [3].

The experimental viscosity data have already been compared by the authors of Refs. [1–3] with *ab initio* theoretically computed viscosity values. This concerns the theoretical values which were calculated by one of us (R. H.), alone or together with different colleagues, for the pure gases nitrogen [4], carbon dioxide [5], methane [6], and ethane [7] as well as for the binary mixtures carbon dioxide–nitrogen [8] and ethane–methane [9]. In addition, correlations for the viscosity values in the limit of zero density were proposed by Laesecke and Muzny for carbon dioxide [10] and methane [11, 12]. These correlations were generated by means of symbolic regression using the theoretically calculated viscosity values of Hellmann for these gases.

However, these mixture data were not employed to determine values for the binary diffusion coefficient. In the past, this was frequently done using experimental viscosity data of binary mixtures, although, as we know today, not in the right manner. The problem is that a ratio of two effective cross-sections for viscosity and diffusion, A_{12}^* , is additionally needed in a twofold way. At first, A_{12}^* is required to deduce the interaction viscosity in the limit of zero density, $\eta_{12}^{(0)}$. For this step, it is not significant whether the ratio is computed using a spherical or a nonspherical potential for the interaction of the molecules. But for the second step, to compute the diffusion coefficient or, more precisely, the product of molar density and diffusion coefficient in the limit of zero density, $(\rho D_{12})^{(0)}$, it is highly significant what this ratio A_{12}^* is calculated for. In the case of a nonspherical potential, A_{12}^* and also $(\rho D_{12})^{(0)}$ are several percent higher. This is in agreement with theoretically computed values of the diffusion coefficients for dilute gas mixtures as can be seen in Ref. [13] for the carbon dioxide–ethane binary gaseous mixture applying the diffusion coefficients of Abe *et al.* [14] derived from viscosity measurements on this mixture. Since the nonspherical potentials for both intermolecular interactions are available by the work of Hellmann *et al.* [8, 9], it is reasonable to also prove this for the binary gaseous mixtures under discussion.

In a recent publication of one of us (E. V.) [15], the results of viscosity measurements on eight gaseous and vapor mixtures were re-evaluated and particularly analyzed to determine the interaction viscosity, $\eta_{12}^{(0)}$, and subsequently to derive the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, both in the limit of zero density. Three procedures were used in Ref. [15] to deduce $\eta_{12}^{(0)}$ applying the information on the measured viscosities for the eight gaseous and vapor mixtures. The first procedure could be used for the gaseous mixtures carbon dioxide–ethane [16] and propane–isobutane [17] as well as for the vapor mixture methanol–triethylamine [18]. In this procedure, density series for only one

unique mole fraction are required for $\eta_{\text{mix}}^{(0)}(T)$ and $\eta_{\text{mix}}^{(1)}(T)$ to be derived. For the vapor mixture methanol–triethylamine, viscosity values at a unique mole fraction were especially created from five of six isochoric series of measurements at nearly the same mole fraction. The values of $\eta_{\text{mix}}^{(0)}(T)$ were employed together with newly determined values $\eta_i^{(0)}$ ($i = 1, 2$) of the pure gases at the temperatures of quasi-isotherms for the mixtures. The root of a quadratic relationship resulting from the Chapman–Enskog theory of dilute gases in its first-order approximation [19] yielded the required $\eta_{12}^{(0)}$ values. A_{12}^* appearing in this relationship was also needed. However, a nonspherical potential was only available for the gaseous mixture carbon dioxide–ethane, so that A_{12}^* had to be computed for the other two mixtures using the effective spherical potential of the interaction of the molecules. But the nonspherical potential for the interaction of the molecules is to be preferred.

In addition, a further scheme to determine $\eta_{12}^{(0)}$ could be applied using a modification of the Enskog theory of hard spheres based on the initial density viscosity coefficients for the mixtures and for the pure gases, $\eta_{\text{mix}}^{(1)}(T)$ and $\eta_i^{(1)}(T)$. This procedure implies that the uncertainty of the initial density viscosity coefficients is very low. In this procedure, a quadratic relationship is also generated, whose root provides the interaction viscosity in the limit of zero density. However, in Ref. [15], the uncertainties of $\eta_i^{(1)}$ of the pure gases and vapors as well as of $\eta_{\text{mix}}^{(1)}(T)$ for the particular mixtures were too large, so that reasonable results for $\eta_{12}^{(0)}(T)$ could not be deduced. In this paper, we will try to derive $\eta_{12}^{(0)}(T)$ by means of $\eta_{\text{mix}}^{(1)}(T)$ and $\eta_i^{(1)}(T)$ assuming that their uncertainties are lower than those of Ref. [15].

The second and third procedures used in Ref. [15] to determine experimentally based $\eta_{12}^{(0)}(T)$ values for the six binary vapor mixtures methanol with triethylamine (all three procedures were applied to this system), benzene, and cyclohexane as well as benzene with toluene, *p*-xylene, and phenol shall not be discussed in detail here.

The results of the measurements at Ruhr University Bochum are suitable for applying at least the first procedure to deduce $\eta_{12}^{(0)}(T)$ and $(\rho D_{12})^{(0)}$ values, since the viscosity data were determined as density series for the pure gases and at constant mole fractions y for the gaseous mixtures.

2 Results of the Viscosity Measurements at Ruhr University Bochum

The dynamic viscosity was measured in Bochum relative to helium, for which the most accurate viscosity values of the dilute gas (in the limit of zero density) were used. These values resulted from ab initio calculations with a relative standard uncertainty of 0.001 % ($k = 1$) [20]. Moreover, the residual damping, usually measured in vacuum, was calibrated in the complete temperature range from 253.15 K to 473.15 K by means of viscosity values ab initio calculated with relative standard uncertainties of 0.001 % for helium [20], of 0.1 % for neon [21], and of 0.1 % for argon [22]. Moreover, reference values of Berg and Moldover [23] were used at 298.15 K (with relative standard uncertainties of 0.001 % for

helium [20], of 0.031 % for neon, and of 0.027 % for argon). The viscosity of pure carbon dioxide was reported in Ref. [3], whereas the results for the other pure gases were listed in Refs. [1, 2]. The relative expanded combined uncertainty ($k = 2$) in viscosity (including uncertainties of temperature and pressure) for all pure gases and the gaseous mixtures under discussion amounted to from 0.14 % to 0.49 %, depending on temperature, with the lower values corresponding to 298.15 K [1–3]. These values of the uncertainties indicate the measurements as very accurate ones.

The isothermal measurements at Ruhr University Bochum were taken up to 2 MPa, meaning that the density series include, at least, a linear density contribution or that even a quadratic member in the density series could appear. The density series for the pure gases and for the gaseous mixtures at a constant mole fraction enabled to determine apart from the viscosities in the limit of zero density, $\eta_i^{(0)}$ and $\eta_{\text{mix}}^{(0)}$, respectively, in any case the viscosity coefficients of the initial density dependence, $\eta_i^{(1)}$ and $\eta_{\text{mix}}^{(1)}$, respectively. The authors of Refs. [1, 2] themselves evaluated the isothermal viscosity measurements dependent on the density for the three pure gases nitrogen, ethane, and methane as well as for the gaseous mixtures of carbon dioxide–nitrogen and ethane–methane with density series restricted to the first (N_2 and $\text{CO}_2\text{--N}_2$) or second (C_2H_6 , CH_4 , and $\text{C}_2\text{H}_6\text{--CH}_4$) power,

$$\eta_i(T, \rho) = \eta_i^{(0)}(T) + \eta_i^{(1)}(T)\rho + \eta_i^{(2)}(T)\rho^2, \quad i = 1, 2, \quad (1)$$

$$\eta_{\text{mix}}(T, \rho) = \eta_{\text{mix}}^{(0)}(T) + \eta_{\text{mix}}^{(1)}(T)\rho + \eta_{\text{mix}}^{(2)}(T)\rho^2. \quad (2)$$

They reported in Refs. [1, 2] the obtained values for $\eta_i^{(0)}$, $\eta_i^{(1)}$ [or for the second viscosity virial coefficient $B_{\eta_i} = \eta_i^{(1)}/\eta_i^{(0)}$], $\eta_{\text{mix}}^{(0)}$, and $\eta_{\text{mix}}^{(1)}$, which we employed in this work. However, in the case of carbon dioxide, the $\eta_i^{(0)}$ and $\eta_i^{(1)}$ values were not listed in Ref. [3]. Therefore, we derived these coefficients from the experimental data of that report. For this, Eq. 1 was limited to the first order in density.

For our purpose, to determine the interaction viscosity, $\eta_{12}^{(0)}(T)$, and the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, each in the limit of zero density, isotherms should be regarded or, at least, quasi-isothermal data should be generated for the viscosities $\eta_i^{(0)}(T)$ of the pure gases and of the respective values $\eta_{\text{mix}}^{(0)}(T)$ of the gaseous mixtures as well as for the initial density viscosity coefficients $\eta_i^{(1)}(T)$ of the pure gases and $\eta_{\text{mix}}^{(1)}(T)$ of the respective gaseous mixtures. However, the experimental data of the pure gases and of the mixtures for a specified mole fraction were not determined at exactly the same temperatures. To create quasi-isothermal viscosity values, the experimental data of the series of measurements on the pure gases and on the mixtures of carbon dioxide–nitrogen and ethane–methane are most suitable, since they are distinguished by only small temperature differences in adjacent measurement points. They were arranged to isotherms, meaning that quasi-isotherms were created for the three mole fractions of the mixtures and for the pure gases in a body. In the following, the different

viscosities are all labeled as $\eta(T)$. They were converted into quasi-isothermal values using a first-order Taylor series in temperature,

$$\eta(T_{\text{iso}}) = \eta(T_{\text{exp}}) + \left(\frac{\partial \eta}{\partial T} \right)_{\rho} (T_{\text{iso}} - T_{\text{exp}}) + R_N. \quad (3)$$

The temperature of the isotherms T_{iso} conforms to the mean of the experimental temperatures T_{exp} of the available data points for the pure gases and for the gaseous mixtures measured at the specified mole fraction. It was checked that the remainder R_N is negligible compared to the experimental uncertainty. The temperature coefficient $(\partial \eta / \partial T)_{\rho}$ needed in Eq. 3 was deduced from the following relationship,

$$\eta(T) = S \exp \left[A \ln(T_R) + \frac{B}{T_R} + \frac{C}{T_R^2} + \frac{D}{T_R^3} + E \right], \quad (4)$$

where $T_R = T / (298.15 \text{ K})$ and $S = 10 \mu\text{Pa} \cdot \text{s}$ or $S = 10 \mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$. The coefficients A , B , C , D , and E were derived from a fit to the experimental data of each series of measurements for the pure gases and for the mixtures at a specified mole fraction. The determined experimental data of the check measurements at approximately 298.15 K of the pure gases each and of the gaseous mixtures at the particular mole fraction were incorporated in the fit. Note that the reason for using a Taylor series instead of employing any fitting function and computing afterwards isothermal viscosity values is the assumption that the experimental errors are kept in this way.

The resulting $\eta_{\text{mix}}^{(0)}$ and $\eta_{\text{mix}}^{(1)}$ values are specified for the three mole fractions y_{N_2} of the gaseous mixture carbon dioxide–nitrogen in Table 1 and for the three mole fractions y_{CH_4} of the gaseous mixture ethane–methane in Table 2. $\eta_i^{(0)}$ and $\eta_i^{(1)}$ values of the pure components at the quasi-isothermal temperatures are also included in the tables.

In order to estimate more precisely the uncertainty of the measurements at Ruhr University Bochum, we calculated for this investigation the viscosity in the limit of zero density for the pure gases and for the three mole fractions at the required temperatures in the third-order approximation of the sophisticated kinetic theory of polyatomic gases [24, 25]. In addition, Crusius *et al.* in Ref. [8] and Hellmann in Ref. [9] recommended for the gaseous mixture carbon dioxide–nitrogen and for the gaseous mixture ethane–methane, respectively, scaled correlations to consider the deviations of their theoretical results for the viscosity of pure carbon dioxide gas, of pure nitrogen gas, and of pure methane gas from the best experimental viscosity data. The viscosity of ethane did not require any scaling. The recommended scaling of the pure gases has been applied to other gaseous mixtures in Refs. [13, 26–30] and is represented for the mixtures under discussion by a temperature-independent factor which depends linearly on the mole fraction:

$$\eta_{\text{rec}}^{(0)} = \eta_{\text{cal}}^{(0)} (1.0055 y_{\text{CO}_2} + 1.0024 y_{\text{N}_2}), \quad (5)$$

Table 1 Values for the viscosity in the limit of zero density, $\eta_{\text{mix}}^{(0)}$, and for the initial density viscosity coefficients, $\eta_{\text{mix}}^{(1)}$, resulting from the quasi-experimental isotherms of the measurements of Humbert *et al.* [1] on the gaseous mixture carbon dioxide–nitrogen including $\eta_i^{(0)}$ and $\eta_i^{(1)}$ of the pure components (data for carbon dioxide from Ref. [3])

T	$\eta_{\text{CO}_2}^{(0)}$	$\eta_{\text{CO}_2}^{(1)}$	$y_{\text{N}_2} = 0.24\,788$		$y_{\text{N}_2} = 0.49\,962$	
			$\eta_{\text{mix}}^{(0)}$	$\eta_{\text{mix}}^{(1)}$	$\eta_{\text{mix}}^{(0)}$	$\eta_{\text{mix}}^{(1)}$
K	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$
298.20	14.8903	0.2090	15.6856	0.3702	16.4677	0.4298
323.21	16.0694	0.2629	16.8915	0.4206	17.6666	0.4346
373.30	18.3844	0.4196	19.2133	0.4938	19.9644	0.4751
423.26	20.6332	0.4461	21.4108	0.5674	22.1302	0.4891
473.30	22.7399	0.5419	23.5203	0.5505	24.2001	0.4550
253.17	12.7132	0.0045	13.4545	0.1883	14.2101	0.3496
273.20	13.6975	0.1005	14.4557	0.3021	15.2283	0.4005
298.21 ^a	14.8928	0.2111	15.6694	0.3714	16.4690	0.4265

T	$y_{\text{N}_2} = 0.74\,981$		$\eta_{\text{N}_2}^{(0)}$	$\eta_{\text{N}_2}^{(1)}$
	$\eta_{\text{mix}}^{(0)}$	$\eta_{\text{mix}}^{(1)}$		
K	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$
298.20	17.1694	0.4309	17.7449	0.3426
323.21	18.3453	0.4458	18.8767	0.3191
373.30	20.5843	0.4512	21.0103	0.3614
423.26	22.6897	0.4175	23.0161	0.3270
473.30	24.6815	0.3826	24.9290	0.2570
253.17	14.9313	0.4196	15.5870	0.4208
273.20	15.9445	0.4369	16.5681	0.4094
298.21 ^a	17.1703	0.4292	17.7382	0.3957

^aCheck measurements

$$\eta_{\text{rec}}^{(0)} = \eta_{\text{cal}}^{(0)} (0.9955 y_{\text{CH}_4} + y_{\text{C}_2\text{H}_6}) . \quad (6)$$

Here $\eta_{\text{rec}}^{(0)}$ and $\eta_{\text{cal}}^{(0)}$ are the recommended and the according to the kinetic theory calculated viscosity values, respectively.

The correlation for the viscosity values in the limit of zero density derived by Laesecke and Muzny for carbon dioxide [10] employing symbolic regression is based on the theoretically calculated viscosity values of Hellmann [5] and includes already the scaling factor 1.0055. Therefore, it is not surprising that the correlated viscosity values of Laesecke and Muzny for pure carbon dioxide are characterized by a tendency to be only smaller by 0.02 % up to 0.04 % than the scaled recommended values of Crusius *et al.* [8] according to Eq. 5.

Table 2 Values for the viscosity in the limit of zero density, $\eta_{\text{mix}}^{(0)}$, and for the initial density viscosity coefficients, $\eta_{\text{mix}}^{(1)}$, resulting from the quasi-experimental isotherms of the measurements of Humberg *et al.* [2] on the gaseous mixture ethane–methane including $\eta_i^{(0)}$ and $\eta_i^{(1)}$ of the pure components

T	$\eta_{\text{C}_2\text{H}_6}^{(0)}$	$\eta_{\text{C}_2\text{H}_6}^{(1)}$	$y_{\text{CH}_4} = 0.2498$		$y_{\text{CH}_4} = 0.4942$	
			$\eta_{\text{mix}}^{(0)}$	$\eta_{\text{mix}}^{(1)}$	$\eta_{\text{mix}}^{(0)}$	$\eta_{\text{mix}}^{(1)}$
K	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$
298.15	9.2260	0.2271	9.6110	0.3390	10.0280	0.3526
323.15	9.9757	0.2563	10.3720	0.3459	10.8060	0.3771
373.15	11.4077	0.4214	11.8280	0.5083	12.2883	0.3696
423.16	12.7882	0.4848	13.2360	0.3441	13.7013	0.3814
473.15	14.1023	0.5557	14.5610	0.4375	15.0420	0.3516
253.15	7.8680	0.0457	8.2250	0.0889	8.5950	0.3607
273.15	8.4770	0.1265	8.8490	0.1360	9.2420	0.2734
298.15 ^a	9.2327	0.2812	9.6110	0.3166	10.0340	0.3056

T	$y_{\text{CH}_4} = 0.7512$		$\eta_{\text{CH}_4}^{(0)}$	$\eta_{\text{CH}_4}^{(1)}$
	$\eta_{\text{mix}}^{(0)}$	$\eta_{\text{mix}}^{(1)}$		
K	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s} \cdot \text{L} \cdot \text{mol}^{-1}$
298.15	10.5267	0.3315	11.0640	0.2310
323.15	11.3130	0.3648	11.8527	0.3339
373.15	12.8153	0.3680	13.3657	0.2945
423.16	14.2413	0.3001	14.7950	0.2936
473.15	15.5883	0.2887	16.1505	0.1877
253.15	9.0620	0.2873	9.5587	0.2924
273.15	9.7240	0.3211	10.2377	0.2820
298.15 ^a	10.5330	0.2724	11.0693	0.3152

^aCheck measurements

In the case of methane, Laesecke and Muzny [11, 12] used guided symbolic regression, too, and the theoretically calculated viscosity values by Hellmann *et al.* [6]. Then in a second step, Laesecke and Muzny adjusted their formulation to the recalibrated experimental data of May *et al.* [31]. In doing so, they determined a factor that leads to the following alternative to Eq. 6:

$$\eta_{\text{rec}}^{(0)} = \eta_{\text{cal}}^{(0)}(0.993388y_{\text{CH}_4} + y_{\text{C}_2\text{H}_6}). \quad (7)$$

The experimentally based viscosity data $\eta_{\text{mix}}^{(0)}(T)$ have already been compared for the gaseous mixtures carbon dioxide–nitrogen and ethane–methane in some figures of Refs. [1, 2, 8] with the theoretically calculated viscosity values $\eta_{\text{cal}}^{(0)}$ and with the recommended ones $\eta_{\text{rec}}^{(0)}(T)$ according to Eqs. 5 and 6, respectively, so that there is no need to repeat it here.

3 Values for the Interaction Viscosity and for the Product of Molar Density and Diffusion Coefficient in the Limit of Zero Density

This report aims at providing reliable values for the limit of zero density of the interaction viscosity, $\eta_{12}^{(0)}(T)$, and of the product of molar density and binary diffusion coefficient, $(\rho D_{12})^{(0)}(T)$. For that purpose, two different procedures dependent on the available information could be used to determine $\eta_{12}^{(0)}(T)$.

We would like to start with the formalism of the second procedure, since the first procedure employs the respective result in the limit of zero density of this second procedure. The measurements on the binary gaseous mixtures carbon dioxide–nitrogen and ethane–methane could be evaluated by means of the linear-in-density expansions of Eq. 2, so that the second procedure could apply the resulting $\eta_{\text{mix}}^{(1)}(T)$ values including the initial density viscosity coefficients $\eta_i^{(1)}(T)$ of the pure gases. This procedure is based on a modification of the Enskog theory of hard-spheres mixtures. Unfortunately, the Rainwater–Friend theory [32–34] describing $\eta_i^{(1)}(T)$ of a pure gas has not been extended to mixtures. Hence, modifications of the Enskog theory for hard-spheres mixtures have to be adopted when representing the initial-density viscosity coefficient of a real dense gaseous mixture. These were described by Di Pippo *et al.* [35], Kestin *et al.* [36], as well as Vesovic and Wakeham [37, 38]. Hendl and Vogel [39–41] developed a variant of this theory in which they proposed to apply the zero-density viscosity values of the pure components and of one mixture at a particular mole fraction to correlate the mixture viscosities at other mole fractions and different moderate densities. This methodology was extended by Humberg *et al.* [1, 2] to correlate the results of their viscosity measurements at temperatures from 253 K to 473 K up to 2 MPa on the gaseous mixtures carbon dioxide–nitrogen and ethane–methane.

In the Enskog theory, the viscosity of a binary hard-spheres mixture can be represented in the measured range of moderate densities considering terms linear in density by

$$\eta_{\text{mix}} = - \left| \begin{array}{ccc} H_{11} & H_{12} & Y_1 \\ H_{21} & H_{22} & Y_2 \\ Y_1 & Y_2 & 0 \end{array} \right| \bigg/ \left| \begin{array}{cc} H_{11} & H_{12} \\ H_{21} & H_{22} \end{array} \right|, \quad (8)$$

which, taking into account that $H_{12} \equiv H_{21}$, leads to

$$\eta_{\text{mix}} = \frac{Y_1^2 H_{22} + Y_2^2 H_{11} - 2 Y_1 Y_2 H_{12}}{H_{11} H_{22} - H_{12}^2}. \quad (9)$$

The quantities H_{ij} , Y_i , and a contact value of the radial distribution function χ_{ij} are given by density series:

$$H_{ij} = H_{ij}^{(0)} + H_{ij}^{(1)} \rho, \quad (10)$$

$$Y_i = y_i + y_i^{(1)} \rho, \quad (11)$$

$$\chi_{ij} = 1 + \chi_{ij}^{(1)} \rho. \quad (12)$$

After inserting Eqs. 10 and 11 into Eq. 9, expanding into a density series, neglecting all density terms beyond linear ones, and comparing with the coefficients of the density series of Eq. 2, one obtains

$$\eta_{\text{mix}}^{(0)} = \left[y_1^2 H_{22}^{(0)} + y_2^2 H_{11}^{(0)} - 2 y_1 y_2 H_{12}^{(0)} \right] \left[H_{11}^{(0)} H_{22}^{(0)} - H_{12}^{(0)2} \right]^{-1}, \quad (13)$$

$$\begin{aligned} \eta_{\text{mix}}^{(1)} = & \left[y_1^2 H_{22}^{(1)} + y_2^2 H_{11}^{(1)} - 2 y_1 y_2 H_{12}^{(1)} + 2 y_1 y_1^{(1)} H_{22}^{(0)} \right. \\ & + 2 y_2 y_2^{(1)} H_{11}^{(0)} - 2 y_1 y_2^{(1)} H_{12}^{(0)} - 2 y_2 y_1^{(1)} H_{12}^{(0)} \\ & \left. - \eta_{\text{mix}}^{(0)} \left(H_{11}^{(0)} H_{22}^{(1)} + H_{11}^{(1)} H_{22}^{(0)} - 2 H_{12}^{(0)} H_{12}^{(1)} \right) \right] \left[H_{11}^{(0)} H_{22}^{(0)} - H_{12}^{(0)2} \right]^{-1}. \end{aligned} \quad (14)$$

The elements of the density series of Eqs. 10–12 are defined as:

$$H_{ii}^{(0)} (i = 1, 2) = \frac{y_i^2}{\eta_i^{(0)}} + \sum_{j=1, j \neq i}^2 \frac{2 y_i y_j}{\eta_{ij}^{(0)}} \frac{m_i m_j}{(m_i + m_j)^2} \left(\frac{5}{3 A_{ij}^*} + \frac{m_j}{m_i} \right), \quad (15)$$

$$H_{ii}^{(1)} (i = 1, 2) = \frac{y_i^2 \chi_{ii}^{(1)}}{\eta_i^{(0)}} + \sum_{j=1, j \neq i}^2 \frac{2 y_i y_j \chi_{ij}^{(1)}}{\eta_{ij}^{(0)}} \frac{m_i m_j}{(m_i + m_j)^2} \left(\frac{5}{3 A_{ij}^*} + \frac{m_j}{m_i} \right), \quad (16)$$

$$H_{ij}^{(0)} (j \neq i) = - \frac{2 y_i y_j}{\eta_{ij}^{(0)}} \frac{m_i m_j}{(m_i + m_j)^2} \left(\frac{5}{3 A_{ij}^*} - 1 \right), \quad (17)$$

$$H_{ij}^{(1)} (j \neq i) = - \frac{2 y_i y_j \chi_{ij}^{(1)}}{\eta_{ij}^{(0)}} \frac{m_i m_j}{(m_i + m_j)^2} \left(\frac{5}{3 A_{ij}^*} - 1 \right), \quad (18)$$

$$\chi_{ij}^{(1)} = \zeta_3 + \frac{3 \sigma_{ii} \sigma_{jj}}{2 \sigma_{ij}} \zeta_2, \quad (19)$$

$$\zeta_k = \frac{\pi}{6} N_A \sum_{i=1}^2 y_i \sigma_{ii}^k, \quad (20)$$

$$y_i^{(1)} (i = 1, 2) = \frac{4}{5} y_i \sum_{j=1}^2 \frac{m_j}{m_i + m_j} y_j b_{ij}, \quad (21)$$

$$b_{12} = \frac{2}{3} \pi N_A \sigma_{12}^3, \quad (22)$$

where b_{12} is the cross second virial coefficient.

Equations 13, 15, and 17 correspond to the well-established kinetic theory of dilute monatomic gases and gaseous mixtures by Chapman and Enskog [19] in its first-order approximation for representing the viscosity of a two-component gaseous mixture in the limit of zero density and form the basis of the first procedure to derive $\eta_{12}^{(0)}$. Here, $y_2 = 1 - y_1$ is the mole fraction of the second component, m_1 and m_2 are the molecular masses of components 1 and 2, whereas $\eta_1^{(0)}$ and $\eta_2^{(0)}$ are the zero-density viscosities of the pure components. In this procedure, the interaction viscosity $\eta_{12}^{(0)}$ corresponds to the viscosity of a hypothetical one-component gas with the molecular mass m_{12} and the interaction potential of the unlike particles and is given as

$$\eta_{12}^{(0)}(T) = \frac{(\pi m_{12} k_B T)^{1/2}}{4\pi \sigma_{12}^2 \mathfrak{G}_\eta^*(T_{12}^*)}, \quad (23)$$

$$m_{12} = \frac{2m_1 m_2}{m_1 + m_2}, \quad (24)$$

$$T_{12}^* = \frac{k_B T}{\varepsilon_{12}}. \quad (25)$$

T_{12}^* is the reduced temperature, while $\mathfrak{G}_\eta^*$ represents a reduced effective cross-section, which considers all the dynamic and statistical information about the binary collisions and, if the first-order kinetic theory framework is applied to real experimental data, incorporates in an effective manner contributions of higher-order kinetic theory approximations. In principle, the scaling parameters ε_{12} and σ_{12} can be derived from those of the pure components by means of the usual approximate mixing rules,

$$\sigma_{12} = \frac{\sigma_1 + \sigma_2}{2}, \quad (26)$$

$$\varepsilon_{12} = (\varepsilon_1 \varepsilon_2)^{1/2}. \quad (27)$$

The scaling parameters for the pure components can be obtained *via* the viscosity of a pure gas i in the limit of zero density according to the kinetic theory of dilute monatomic gases by Chapman and Enskog [19],

$$\eta_i^{(0)}(T) = \frac{(\pi m_i k_B T)^{1/2}}{4\pi \sigma_i^2 \mathfrak{G}_\eta^*(T_i^*)}. \quad (28)$$

Then, the analysis of the experimental $\eta_i^{(0)}$ values is based on the extended corresponding states principle within the kinetic theory of monatomic gases, in which the temperature dependence of the reduced effective cross-section $\mathfrak{S}_\eta^*(T_i^*)$ is represented by the following functional form,

$$\ln \mathfrak{S}_\eta^*(T_i^*) = \sum_{k=0}^4 a_k (\ln T_i^*)^k. \quad (29)$$

In a first step, a universal correlation using the coefficients a_k of the functional $\mathfrak{S}_\eta^*$ for the rare gases published by Bich *et al.* [42] is applied to derive the scaling factors σ_i and ϵ_i/k_B . After that, a second step follows typically, in which an individual correlation is performed and new coefficients a_k are deduced by fitting Eqs. 28 and 29 to the experimental $\eta_i^{(0)}$ values. This second step is not necessary if, as in our case, the scaling factors ϵ_i and σ_i are used in Eqs. 26 and 27 to compute the scaling parameters σ_{12} and ϵ_{12} . The reason is that the reduced effective cross-section $\mathfrak{S}_\eta^*(T_{12}^*)$ in Eq. 23 is calculated by using the coefficients a_k for a universal correlation.

A_{12}^* is a temperature dependent ratio of two effective cross-sections for the viscosity (enumerator) and diffusion (denominator),

$$A_{12}^* = \frac{5}{6} \frac{\mathfrak{S}(2000)_{12}}{\mathfrak{S}'(1000)_{12}}. \quad (30)$$

It is often approximated by a constant value of 1.10. In the framework of the corresponding states principle, A_{12}^* can be determined more accurately as a function of temperature for a spherical potential of the intermolecular interaction using universal correlations for the needed two cross-sections [42]. However, for molecules it is more appropriate to compute this ratio applying the nonspherical potential of the intermolecular interaction.

The first procedure was applied for both the carbon dioxide–nitrogen and ethane–methane mixtures. In doing so, the values of the zero-density viscosity values $\eta_{\text{mix}}^{(0)}(T)$ of the mixtures for a specified mole fraction y_2 as well as $\eta_i^{(0)}$ ($i = 1, 2$) of the pure gases at the temperatures of the quasi-isotherms of Tables 1 and 2 were used. A rearrangement of Eqs. 13, 15, and 17 leads to a quadratic relationship, the root of which yields the required interaction viscosity. In a first step of the procedure, A_{12}^* values as a function of temperature following from the corresponding states principle by means of universal correlations for the needed two cross-sections [42] were used. These cross-sections are based on an effective spherical potential. The results of $\eta_{12}^{(0)}$ for the three mole fractions each of the carbon dioxide–nitrogen and of the ethane–methane mixtures including the average together with the applied A_{12}^* values are given in Tables 3 and 4. For a second step, we computed improved A_{12}^* values for the nonspherical carbon dioxide–nitrogen and ethane–methane potentials. This second step is motivated by findings described in Ref. [43]. In Fig. 7.9a of this reference, it is demonstrated that A_{12}^* values of molecules are distinctly increased by several percent if their nonspherical potentials are used instead of an effective spherical potential. The results for the improved A_{12}^* values computed applying the nonspherical potentials of the intermolecular interaction of carbon dioxide and nitrogen [8] and of ethane and methane [9] are

Table 3 Values for the limit of zero density of the interaction viscosity, $\eta_{12}^{(0)}$, and of the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, resulting from the $\eta_{\text{mix}}^{(0)}$ and $\eta_i^{(0)}$ ($i = 1, 2$) values of Table 1 for the measurements of Humberg *et al.* [1] and using A_{12}^* values from the effective spherical potential for the gaseous mixture carbon dioxide–nitrogen

T	$y_{\text{N}_2} =$ $A_{12}^*{}^a$	0.24 788 $\eta_{12}^{(0)}$	0.49 962 $\eta_{12}^{(0)}$	0.74 981 $\eta_{12}^{(0)}$	Average $\eta_{12}^{(0)}$	$10^4(\rho D_{12})^{(0)}$
K		$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\text{mol} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$
298.20	1.1041	16.635	16.638	16.626	16.633	6.437
323.21	1.1038	17.873	17.847	17.826	17.849	6.906
373.30	1.1038	20.173	20.143	20.124	20.147	7.795
423.26	1.1045	22.237	22.268	22.271	22.259	8.617
473.30	1.1055	24.329	24.331	24.292	24.317	9.423
253.17	1.1054	14.345	14.344	14.334	14.341	5.557
273.20	1.1047	15.355	15.373	15.367	15.365	5.950
298.21 ^b	1.1041	16.578	16.642	16.636	16.619	6.431

^aValues for A_{12}^* were determined using the corresponding states principle for spherical molecules [42]

^bCheck measurements

Table 4 Values for the limit of zero density of the interaction viscosity, $\eta_{12}^{(0)}$, and of the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, resulting from the $\eta_{\text{mix}}^{(0)}$ and $\eta_i^{(0)}$ ($i = 1, 2$) values of Table 2 for the measurements of Humberg *et al.* [2] and using A_{12}^* values from the effective spherical potential for the gaseous mixture ethane–methane

T	$y_{\text{CH}_4} =$ $A_{12}^*{}^a$	0.2498 $\eta_{12}^{(0)}$	0.4942 $\eta_{12}^{(0)}$	0.7512 $\eta_{12}^{(0)}$	Average $\eta_{12}^{(0)}$	$10^4(\rho D_{12})^{(0)}$
K		$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\text{mol} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$
298.15	1.1067	9.832	9.824	9.830	9.829	6.238
323.15	1.1056	10.580	10.585	10.589	10.585	6.712
373.15	1.1043	12.017	12.030	12.034	12.027	7.618
423.16	1.1037	13.420	13.402	13.409	13.410	8.489
473.15	1.1037	14.708	14.701	14.700	14.703	9.308
253.15	1.1094	8.461	8.423	8.445	8.443	5.372
273.15	1.1080	9.083	9.059	9.073	9.072	5.765
298.15 ^b	1.1067	9.817	9.830	9.837	9.828	6.238

^aValues for A_{12}^* were determined using the corresponding states principle for spherical molecules [42]

^bCheck measurements

listed in Tables 5 and 6. In addition, values of the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, are also provided in these tables. The interaction viscosity $\eta_{12}^{(0)}$ is related to that product according to the first-order approximation of the kinetic theory of dilute gases as

Table 5 Values for the limit of zero density of the interaction viscosity, $\eta_{12}^{(0)}$, and of the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, resulting from the $\eta_{\text{mix}}^{(0)}$ and $\eta_i^{(0)}$ ($i = 1, 2$) values of Table 1 for the measurements of Humberg *et al.* [1] and using A_{12}^* values from the nonspherical potential for the gaseous mixture carbon dioxide–nitrogen

T	$y_{\text{N}_2} =$ $A_{12}^*{}^a$	0.24 788 $\eta_{12}^{(0)}$	0.49 962 $\eta_{12}^{(0)}$	0.74 981 $\eta_{12}^{(0)}$	Average $\eta_{12}^{(0)}$	$10^4(\rho D_{12})^{(0)}$
K		$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\text{mol} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$
298.20	1.1708	16.632	16.634	16.621	16.629	6.824
323.21	1.1714	17.870	17.842	17.820	17.844	7.327
373.30	1.1719	20.168	20.137	20.115	20.140	8.273
423.26	1.1719	22.231	22.259	22.260	22.250	9.140
473.30	1.1717	24.321	24.321	24.279	24.307	9.983
253.17	1.1679	14.343	14.342	14.331	14.338	5.870
273.20	1.1695	15.353	15.370	15.362	15.362	6.297
298.21 ^b	1.1707	16.576	16.638	16.630	16.614	6.818

^aValues for A_{12}^* were calculated according to Crusius *et al.* [8] for the nonspherical potential

^bCheck measurements

Table 6 Values for the limit of zero density of the interaction viscosity, $\eta_{12}^{(0)}$, and of the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, resulting from the $\eta_{\text{mix}}^{(0)}$ and $\eta_i^{(0)}$ ($i = 1, 2$) values of Table 2 for the measurements of Humberg *et al.* [2] and using A_{12}^* values from the nonspherical potential for the gaseous mixture ethane–methane

T	$y_{\text{CH}_4} =$ $A_{12}^*{}^a$	0.2498 $\eta_{12}^{(0)}$	0.4942 $\eta_{12}^{(0)}$	0.7512 $\eta_{12}^{(0)}$	Average $\eta_{12}^{(0)}$	$10^4(\rho D_{12})^{(0)}$
K		$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\text{mol} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$
298.15	1.1411	9.828	9.821	9.827	9.825	6.430
323.15	1.1422	10.576	10.581	10.585	10.581	6.931
373.15	1.1430	12.012	12.025	12.029	12.022	7.881
423.16	1.1430	13.414	13.396	13.402	13.404	8.788
473.15	1.1428	14.702	14.694	14.692	14.696	9.632
253.15	1.1369	8.459	8.421	8.442	8.441	5.504
273.15	1.1392	9.081	9.056	9.070	9.069	5.926
298.15 ^b	1.1411	9.814	9.826	9.834	9.824	6.430

^aValues for A_{12}^* were calculated according to Hellmann [9] for the nonspherical potential

^bCheck measurements

$$(\rho D_{12})^{(0)} = \frac{3}{5} \frac{m_1 + m_2}{m_1 m_2} \frac{1}{N_A} A_{12}^* \eta_{12}^{(0)}, \quad (31)$$

where N_A is Avogadro's constant. To gain experimentally based values of $(\rho D_{12,\text{exp}})^{(0)}$ by means of Eq. 31, the values for A_{12}^* were those chosen in Tables 3

and 4, respectively, as well as selected in Tables 5 and 6, respectively, i.e., those derived for the respective effective spherical potential and for the particular realistic nonspherical potentials each.

Equations 9–22 are usually applied as correlation and extrapolation schemes for the composition and temperature dependencies of the viscosity of moderately dense gaseous and vapor mixtures. These schemes usually include an additional modification of the Enskog theory (see below). In principle, Eq. 14 together with Eqs. 13 and 15–22 can be used to generate a quadratic relationship, whose root provides the interaction viscosity in the limit of zero density.

This second procedure uses the idea of the interaction viscosity for hard spheres $\eta_{12}^{(0)}$ according to

$$\eta_{12}^{(0)} = \frac{5}{16} \frac{(\pi m_{12} k_B T)^{1/2}}{\pi \sigma_{12}^2}, \quad (32)$$

where m_{12} and σ_{12} correspond to those of Eqs. 24 and 26, respectively. σ_i are the diameters of the hard spheres. In addition, one needs the formalism for a pure hard-sphere gas,

$$\eta_i^{(0)} = \frac{5}{16} \frac{(\pi m_i k_B T)^{1/2}}{\pi \sigma_i^2}, \quad (33)$$

$$\eta_i^{(1)} = \eta_i^{(0)} \left(\frac{4}{5} b_i - \chi_i^{(1)} \right), \quad (34)$$

$$b_i = \frac{2}{3} \pi N_A \sigma_i^3, \quad (35)$$

$$\chi_i = 1 + \chi_i^{(1)} \rho, \quad (36)$$

$$\chi_i^{(1)} = \frac{5}{12} \pi N_A \sigma_i^3. \quad (37)$$

Here, b_i is the second virial coefficient and χ_i is the equilibrium radial distribution function at contact, each for hard spheres.

In the adaptation of the Enskog theory of hard spheres to real molecules, it is assumed that the transport coefficients for real molecules have the same form as in the case of a hard-spheres gas. Then, the hard-spheres quantities can be replaced by real-gas quantities. This leads to the distinction among three real-gas diameters: σ_i^η for $\eta_i^{(0)}$, σ_i^b for b_i , and σ_i^χ for $\chi_i^{(1)}$. The main modification consists in the pressure p in the equation of state for hard spheres being replaced by the thermal pressure $T(\partial p / \partial T)_\rho$ of the real gas, which means

$$b_i = B_i + T \frac{dB_i}{dT}, \quad b_{12} = B_{12} + T \frac{dB_{12}}{dT}, \quad (38)$$

where B_i is the second virial coefficient of the real gas and B_{12} is the cross second virial coefficient of the real gaseous mixture. The B_i and B_{12} values for the considered gases as well as for the respective gaseous mixtures were taken from Ref. [44]. Then, σ_i^b and σ_{12}^b can be determined directly from $B_i(T)$ as well as $B_{12}(T)$ data by means of

$$\sigma_i^b = \left(\frac{3}{2\pi N_A} b_i \right)^{1/3}, \quad \sigma_{12}^b = \left(\frac{3}{2\pi N_A} b_{12} \right)^{1/3}. \quad (39)$$

Finally, σ_i^χ follows from Eqs. 34 and 37 with b_i , whereas σ_i^η results from $\eta_i^{(0)}$. The scaling factors σ_i for the pure components as well as σ_{12} for the mixtures are needed in connection with the hard-spheres modification.

The results averaged for the individual mole fractions of this second procedure using $\eta_i^{(1)}(T)$ of the pure components as well as $\eta_{\text{mix}}^{(1)}(T)$ of the gaseous mixtures are presented in Table 7 applying the A_{12}^* values calculated by us for the non-spherical potentials. The outcome using the A_{12}^* values computed for the spherical potentials is not given here, since the uncertainties of the initial density viscosity coefficients $\eta_i^{(1)}$ of the pure gases as well as of $\eta_{\text{mix}}^{(1)}(T)$ for the particular mixtures are comparatively large, so that the results for the individual mole fractions

Table 7 Averaged values for the limit of zero density of the interaction viscosity, $\eta_{12}^{(0)}$, and of the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$, resulting from the $\eta_{\text{mix}}^{(1)}$ and $\eta_i^{(1)}$ values of Table 1 for the measurements of Humberg *et al.* [1] as well as for the gaseous mixture ethane–methane resulting from the $\eta_{\text{mix}}^{(1)}$ and $\eta_i^{(1)}$ values of Table 2 for the measurements of Humberg *et al.* [2], in each case using A_{12}^* values from the nonspherical potential

Carbon dioxide–nitrogen				Ethane–methane			
T	A_{12}^* ^a	$\eta_{12}^{(0)}$	$10^4(\rho D_{12})^{(0)}$	T	A_{12}^* ^b	$\eta_{12}^{(0)}$	$10^4(\rho D_{12})^{(0)}$
K		$\mu\text{Pa} \cdot \text{s}$	$\text{mol} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$	K		$\mu\text{Pa} \cdot \text{s}$	$\text{mol} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$
298.20	1.1708	22.620	9.283	298.15	1.1411	11.841	7.749
323.21	1.1714	23.853	9.794	323.15	1.1422	11.917	7.807
373.30	1.1719	24.007	9.862	373.15	1.1430	13.789	9.040
423.26	1.1719	26.768	10.995	423.16	1.1430	13.037	8.547
473.30	1.1717	27.714	11.382	473.15	1.1428	15.435	10.117
253.17	1.1679	19.970	8.175	253.15	1.1369	9.469	6.174
273.20	1.1695	21.209	8.694	273.15	1.1392	9.815	6.413
298.21 ^c	1.1707	21.718	8.912	298.15 ³	1.1411	10.026	6.562

^aValues for A_{12}^* were calculated according to Crusius *et al.* [8] for the nonspherical potential

^bValues for A_{12}^* were calculated according to Hellmann [9] for the nonspherical potential

^cCheck measurements

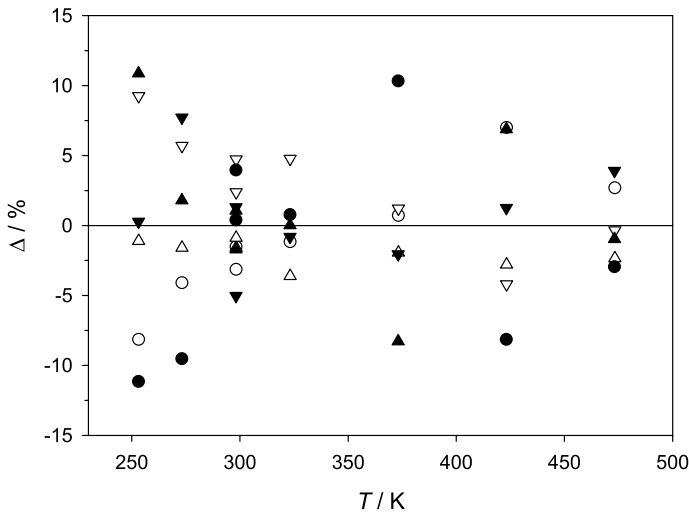


Fig. 1 Relative differences of the interaction viscosity values $\eta_{12,2nd\text{ proc},mol}^{(0)}(T)$ of the second procedure for the individual mole fractions of the gaseous mixtures carbon dioxide–nitrogen and ethane–methane from the averaged interaction viscosity values $\eta_{12,2nd\text{ proc},aver}^{(0)}(T)$, as a function of temperature T . Deviations: $\Delta = 100 \cdot (\eta_{12,2nd\text{ proc},mol}^{(0)} - \eta_{12,2nd\text{ proc},aver}^{(0)}) / \eta_{12,2nd\text{ proc},aver}^{(0)}$. Carbon dioxide–nitrogen: $\circ$, $y_{N_2} = 0.24788$; Δ , $y_{N_2} = 0.49962$; ∇ , $y_{N_2} = 0.74981$. Ethane–methane: $\bullet$, $y_{CH_4} = 0.2498$; $\blacktriangle$, $y_{CH_4} = 0.4942$; $\blacktriangledown$, $y_{CH_4} = 0.7512$

illustrated as relative differences from the averaged interaction viscosity values $\eta_{12,2nd\text{ proc},aver}^{(0)}$ in Fig. 1 show substantial deviations.

4 Discussion

The discussion of the results for the interaction viscosity $\eta_{12}^{(0)}$ is started with the second procedure, which applies the initial density viscosity coefficients $\eta_i^{(1)}$ and $\eta_{mix}^{(1)}$. At the end of the previous section, it has already been stated that the uncertainties of the initial density viscosity coefficients are serious.

Without anticipating the results of the first procedure, it is significant that $\eta_{12}^{(0)}$ following from $\eta_i^{(0)}$ and $\eta_{mix}^{(0)}$ amount to lower values. This is illustrated in Fig. 2, in which $\eta_{12,2nd\text{ proc},aver}^{(0)}(T)$, the averaged interaction viscosity values of the second procedure, are compared with those of the first procedure $\eta_{12,1st\text{ proc},aver}^{(0)}(T)$. The values of the second procedure are for the gaseous system carbon dioxide–nitrogen higher by (14–40) %, whereas those for the mixture ethane–methane are increased by up to 20 %.

Moreover, a recalculation of the viscosity values η_{mix} was performed and in Table 8 compared with the experimental values $\eta_{mix,exp}$ measured by Humberg *et al.* [1, 2] for the gaseous systems under discussion. This comparison is restricted to only one density series for both systems, at the middle mole fraction and at ambient temperature each. Ambient temperature was chosen because the measurements

have the lowest uncertainty under this condition. The results of $\eta_{12}^{(0)}$ for the first and second procedures at the given mole fraction and at the respective temperature are also listed. Three different ways for the recalculation were tested. The third column in the table corresponds to Eq. 2 applying the results of $\eta_{12}^{(0)}$ of the first procedure in the coefficient $\eta_{\text{mix}}^{(0)}$ (Eq. 13) and of $\eta_{12}^{(0)}$ of the second procedure in the coefficient $\eta_{\text{mix}}^{(1)}$ (Eq. 14) of the series expansion. The findings show that the experimental data $\eta_{\text{mix, exp}}$ are consistently described. In the fourth and fifth columns, the findings are reported which followed for the second and third ways if the ratio of determinants (Eq. 8) was computed using either $\eta_{12}^{(0)}$ of the first or of the second procedures. The experimental data $\eta_{\text{mix, exp}}$ are underestimated using $\eta_{12}^{(0)}$ of the first procedure, whereas they are overestimated applying $\eta_{12}^{(0)}$ of the second procedure. Table 8 illustrates that the approximations included in the Enskog theory for binary hard-spheres mixtures at moderate densities are unsuitable to represent the viscosity of a real binary gas mixture. The conclusion is that the initial density viscosity coefficients of the pure gases $\eta_i^{(1)}$ as well as of the gaseous mixture $\eta_{\text{mix}}^{(1)}$ cannot be used to derive reasonable values of $\eta_{12}^{(0)}$.

The results of the interaction viscosity $\eta_{12}^{(0)}$ for the first procedure are summarized for the gaseous mixtures carbon dioxide–nitrogen and ethane–methane in Tables 3, 4, 5, and 6 and should now be investigated. For this, not only the cross-section ratio A_{12}^* but also the interaction viscosity in the limit of zero density, $\eta_{12, \text{cal}}^{(0)}(T)$, were computed for the nonspherical potentials of the intermolecular interaction of the

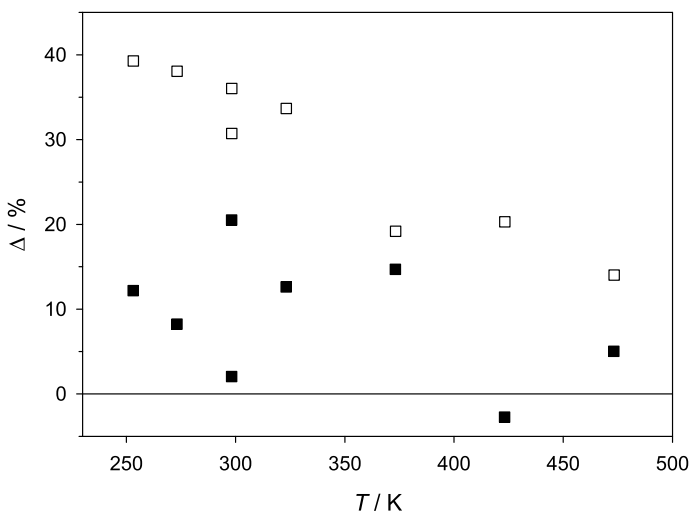


Fig. 2 Relative differences of the averaged interaction viscosity values $\eta_{12, 2\text{nd proc, aver}}^{(0)}(T)$ of the second procedure for the gaseous mixtures carbon dioxide–nitrogen and ethane–methane from the averaged interaction viscosity values $\eta_{12, 1\text{st proc, aver}}^{(0)}(T)$ of the first procedure, as a function of temperature T . Deviations: $\Delta = 100 \cdot (\eta_{12, 2\text{nd proc, aver}}^{(0)} - \eta_{12, 1\text{st proc, aver}}^{(0)}) / \eta_{12, 1\text{st proc, aver}}^{(0)}$. □, carbon dioxide–nitrogen; ■, ethane–methane

Table 8 Recalculated viscosity values, η_{mix} , using the zero-density interaction viscosity values $\eta_{12}^{(0)}$ resulting from the first procedure applying the zero-density viscosities $\eta_i^{(0)}$ and $\eta_{\text{mix}}^{(0)}$ and from the second procedure applying the initial density viscosity coefficients $\eta_i^{(1)}$ and $\eta_{\text{mix}}^{(1)}$, respectively, for the gaseous systems carbon dioxide–nitrogen of the measurements of Humberg *et al.* [1] and ethane–methane of the measurements of Humberg *et al.* [2]

Carbon dioxide–nitrogen				
$T = 298.20 \text{ K}$	$y_{\text{N}_2} = 0.49962$	$\eta_{12}^{(0)} = 16.634 \text{ } \mu\text{Pa} \cdot \text{s}$ from 1st procedure	$\eta_{12}^{(0)} = 22.261 \text{ } \mu\text{Pa} \cdot \text{s}$ from 2nd procedure	
ρ	$\eta_{\text{mix, exp}}$	$\eta_{\text{mix}}^{\text{a}}$, series expansion	$\eta_{\text{mix}}^{\text{b}}$, determ, 1st proc	$\eta_{\text{mix}}^{\text{c}}$, determ, 2nd proc
$\text{mol} \cdot \text{L}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$
0.4977	16.696	16.6816	16.5323	18.8458
0.4122	16.641	16.6449	16.5207	18.8333
0.3287	16.599	16.6090	16.5096	18.8212
0.2425	16.563	16.5719	16.4984	18.8090
0.1624	16.542	16.5375	16.4881	18.7979
0.0829	16.502	16.5033	16.4780	18.7870
0.0437	16.496	16.4865	16.4731	18.7817
Ethane–methane				
$T = 298.15 \text{ K}$	$y_{\text{N}_2} = 0.4942$	$\eta_{12}^{(0)} = 9.821 \text{ } \mu\text{Pa} \cdot \text{s}$ from 1st procedure	$\eta_{12}^{(0)} = 11.638 \text{ } \mu\text{Pa} \cdot \text{s}$ from 2nd procedure	
ρ	$\eta_{\text{mix, exp}}$	$\eta_{\text{mix}}^{\text{a}}$, series expansion	$\eta_{\text{mix}}^{\text{b}}$, determ, 1st proc	$\eta_{\text{mix}}^{\text{c}}$, determ, 2nd proc
$\text{mol} \cdot \text{L}^{-1}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$	$\mu\text{Pa} \cdot \text{s}$
0.4218	10.185	10.1767	10.1162	10.9427
0.3340	10.150	10.1458	10.0976	10.9228
0.2486	10.118	10.1157	10.0796	10.9035
0.1648	10.088	10.0861	10.0621	10.8848
0.0817	10.058	10.0568	10.0448	10.8663
0.0418	10.042	10.0427	10.0366	10.8575

^a $\eta_{12}^{(0)}$ resulting from the first procedure as well as $\eta_{12}^{(0)}$ following from the second procedure were used in the first (Eq. 13) and second (Eq. 14), respectively, members of the series expansion of Eq. 2

^b $\eta_{12}^{(0)}$ resulting from the first procedure was applied in the ratio of determinants of Eq. 8

^c $\eta_{12}^{(0)}$ resulting from the second procedure was used in the ratio of determinants of Eq. 8

systems carbon dioxide–nitrogen and ethane–methane. This could only be done in the first-order approximation of the sophisticated kinetic theory for polyatomic molecules [24, 25], since these quantities are only defined for this approximation.

The average values of the experimentally based interaction viscosity $\eta_{12,\text{exp}}^{(0)}$ given in Table 3 for the mixture carbon dioxide–nitrogen are compared in Fig. 3 with the theoretical values $\eta_{12,\text{cal}}^{(0)}$ computed at the temperatures of this paper. The figure shows that the experimentally based data, applying A_{12}^* values derived from the corresponding states principle using universal correlations for a spherical potential

[42], are roughly between 0.5 % and 0.85 % too high. Some scatter should be caused by experimental uncertainties. Furthermore, the figure illustrates a comparison of the $\eta_{12,\text{cal}}^{(0)}$ values with the experimentally based $\eta_{12,\text{exp}}^{(0)}$ data of Table 5, applying the A_{12}^* values for the nonspherical potential. The figure shows that the agreement is only slightly improved. But one has to take into account that the quantity $\eta_{12,\text{cal}}^{(0)}$ could only be computed in the first-order approximation. Hence, the existing deviations are partly due to the comparison with an effective experimental quantity that only has a counterpart in the kinetic theory of gases at the lowest-order approximation.

The average values of the interaction viscosity $\eta_{12,\text{exp}}^{(0)}$ listed in Table 4 for the mixture ethane–methane are checked in Fig. 4 against the theoretical values $\eta_{12,\text{cal}}^{(0)}$ calculated at the temperatures of this investigation. The figure illustrates in comparison with Fig. 3 that the experimentally based data are better represented for this gaseous mixture, but the relative differences are negative. This concerns both the deviations applying the A_{12}^* values derived from the corresponding states principle using universal correlations for a spherical potential [42], and those resulting with the experimentally based $\eta_{12,\text{exp}}^{(0)}$ data of Table 6 applying the A_{12}^* values for the nonspherical potential. Surprisingly, the $\eta_{12,\text{exp}}^{(0)}$ data derived with the A_{12}^* values for the nonspheri-

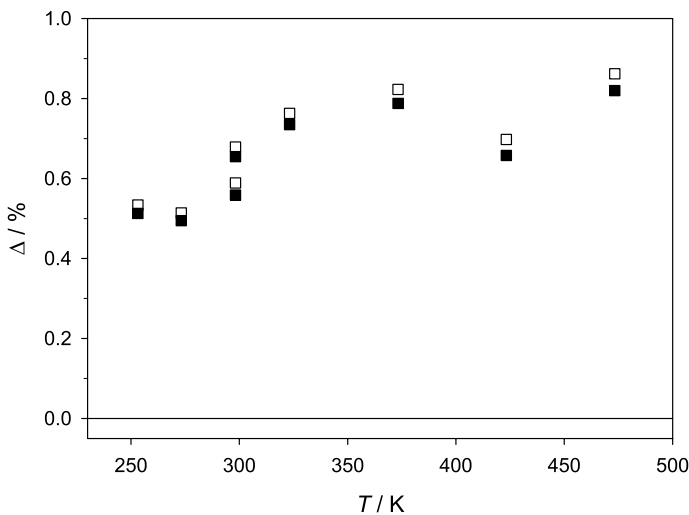


Fig. 3 Comparison of the theoretically calculated values of the interaction viscosity $\eta_{12,\text{cal}}^{(0)}(T)$ for the gaseous mixture carbon dioxide–nitrogen with the average of the experimentally based zero-density viscosity data $\eta_{12,\text{exp}}^{(0)}(T)$ determined by means of Eqs. 13, 15, and 17 using A_{12}^* values derived either from the corresponding states principle for a spherical potential given in Table 3 or from the kinetic theory of polyatomic gases for a nonspherical potential listed in Table 5, as a function of temperature T . Deviations: $\Delta = 100 \cdot (\eta_{12,\text{exp}}^{(0)} - \eta_{12,\text{cal}}^{(0)}) / \eta_{12,\text{cal}}^{(0)}$. □, using A_{12}^* values derived from the corresponding states principle for a spherical potential [42]; ■, using A_{12}^* deduced from the kinetic theory of polyatomic gases for the nonspherical potential [8]

cal potential are less optimally represented by the theoretically computed values $\eta_{12,\text{cal}}^{(0)}$.

The experimentally based data of the product $(\rho D_{12,\text{exp}})^{(0)}$ of Table 3 for the carbon dioxide–nitrogen mixture, applying A_{12}^* values derived from the corresponding states principle using universal correlations for a spherical potential [42], are compared in Fig. 5 with theoretical values $(\rho D_{12,\text{cal}})^{(0)}$, which were again computed for the nonspherical potential of the interaction between the molecules by means of the kinetic theory for polyatomic gases in its third-order approximation [8]. Figure 5 shows that deviations between -4.0% and -5.4% occur. Figure 7 of Ref. [8] illustrates deviations of experimental data $(\rho D_{12,\text{exp}})^{(0)}$ of different authors from theoretical values $(\rho D_{12,\text{cal}})^{(0)}$ computed for the nonspherical potential. Crusius *et al.* [8] found that the data of Robjohns and Dunlop [45] given in the temperature range between 277 K and 323 K agree with their calculations within $\pm 0.2\%$, whereas the data of most other authors are characterized by deviations $> \pm 1\%$. Moreover, Fig. 5 of this paper demonstrates that for the A_{12}^* values computed for the nonspherical potential and applied for Table 5 the experimentally based $(\rho D_{12,\text{exp}})^{(0)}$ data nearly perfectly agree with the computed $(\rho D_{12,\text{cal}})^{(0)}$ values of this paper. The relative differences amount to $+0.33\%$ up to $+0.53\%$.

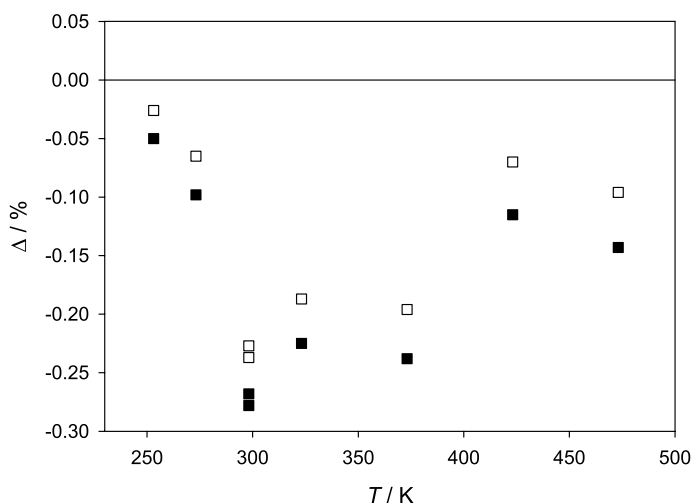


Fig. 4 Comparison of the theoretically calculated values of the interaction viscosity $\eta_{12,\text{cal}}^{(0)}(T)$ for the gaseous mixture ethane–methane with the average of the experimentally based zero-density viscosity data $\eta_{12,\text{exp}}^{(0)}(T)$ determined by means of Eqs. 13, 15, and 17 using A_{12}^* values derived either from the corresponding states principle for a spherical potential given in Table 4 or from the kinetic theory of polyatomic gases for a nonspherical potential listed in Table 6, as a function of temperature T . Deviations: $\Delta = 100 \cdot (\eta_{12,\text{exp}}^{(0)} - \eta_{12,\text{cal}}^{(0)}) / \eta_{12,\text{cal}}^{(0)}$. □, using A_{12}^* values derived from the corresponding states principle for a spherical potential [42]; ■, using A_{12}^* deduced from the kinetic theory of polyatomic gases for the nonspherical potential [9]

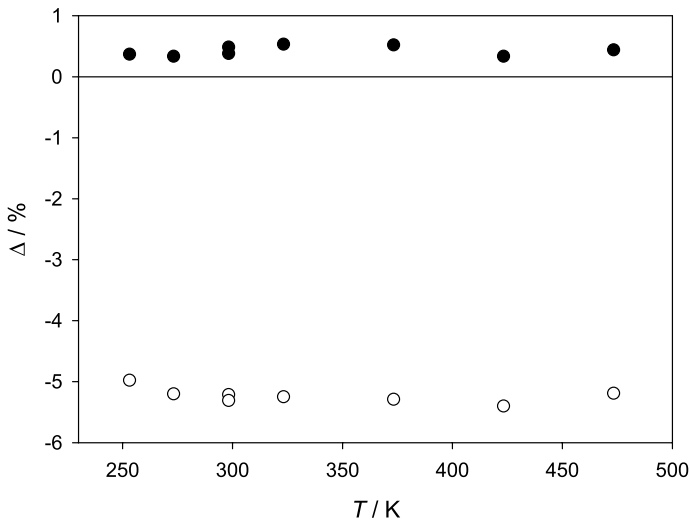


Fig. 5 Comparison of the theoretically calculated values of the product of molar density and binary diffusion coefficient in the limit of zero density $(\rho D_{12,\text{cal}})^{(0)}(T)$ for the gaseous mixture carbon dioxide–nitrogen with experimentally based data $(\rho D_{12,\text{exp}})^{(0)}(T)$, determined by means of Eq. 31 applying $\eta_{12,\text{exp}}^{(0)}(T)$ and A_{12}^* values derived either from the corresponding states principle for a spherical potential given in Table 3 or from the kinetic theory of polyatomic gases for a nonspherical potential listed in Table 5, as a function of temperature T . Deviations: $\Delta = 100 \cdot [(\rho D_{12,\text{exp}})^{(0)} - (\rho D_{12,\text{cal}})^{(0)}] / (\rho D_{12,\text{cal}})^{(0)}$. ○, using A_{12}^* values derived from the corresponding states principle for a spherical potential [42]; ●, using A_{12}^* deduced from the kinetic theory of polyatomic gases for a nonspherical potential [8]

The experimentally based data of the product $(\rho D_{12,\text{exp}})^{(0)}$ of Table 4 for the ethane–methane mixture, applying A_{12}^* values derived from the corresponding states principle using universal correlations for a spherical potential [42], are compared in Fig. 6 with theoretical values $(\rho D_{12,\text{cal}})^{(0)}$, which were again computed for the nonspherical potential of the interaction between the molecules by means of the kinetic theory for polyatomic gases in its third-order approximation [9]. Figure 6 shows that deviations between -2.5% and -3.7% appear. Figure 11 of Ref. [9] illustrates deviations of experimental data $(\rho D_{12,\text{exp}})^{(0)}$ of a few authors from theoretical values $(\rho D_{12,\text{cal}})^{(0)}$ computed for the nonspherical potential. Hellmann [9] found that the data of Arora *et al.* [46] given in the temperature range from 275 K to 323 K differ from his calculations only by about -0.5% , whereas the data of the other authors show deviations $> \pm 1\%$. Moreover, Fig. 6 of this paper demonstrates that for the A_{12}^* values computed for the nonspherical potential and applied for Table 6 the experimentally based $(\rho D_{12,\text{exp}})^{(0)}$ data are lower by 0.05 % up to 0.30 % than the computed $(\rho D_{12,\text{cal}})^{(0)}$ values of this paper.

The conclusion of Ref. [43] (page 243) that inelastic collisions have a fractionally larger effect on the transfer of mass than on the transfer of momentum is justified for the gaseous systems under discussion. This is obvious if the influence of the A_{12}^* values, deduced applying the corresponding states principle with universal correlations for

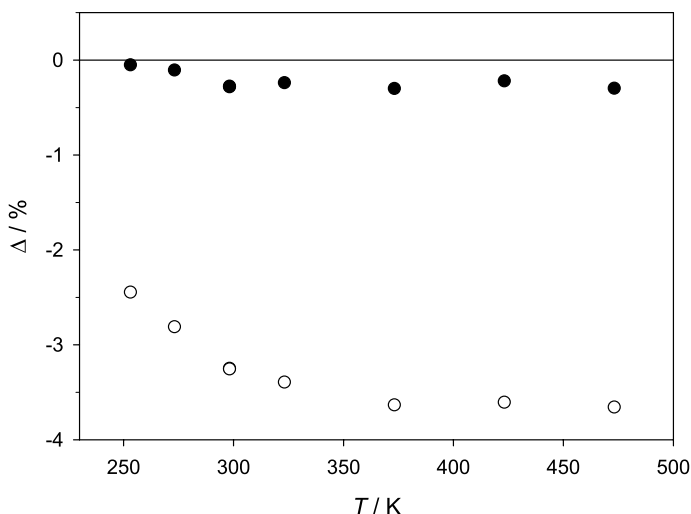


Fig. 6 Comparison of the theoretically calculated values of the product of molar density and binary diffusion coefficient in the limit of zero density $(\rho D_{12,\text{cal}})^{(0)}(T)$ for the gaseous mixture ethane–methane with experimentally based data $(\rho D_{12,\text{exp}})^{(0)}(T)$, determined by means of Eq. 31 applying $\eta_{12,\text{exp}}^{(0)}(T)$ and A_{12}^* values derived either from the corresponding states principle for a spherical potential given in Table 4 or from the kinetic theory of polyatomic gases for a nonspherical potential listed in Table 6, as a function of temperature T . Deviations: $\Delta = 100 \cdot [(\rho D_{12,\text{exp}})^{(0)} - (\rho D_{12,\text{cal}})^{(0)}] / (\rho D_{12,\text{cal}})^{(0)}$. ○, using A_{12}^* values derived from the corresponding states principle for a spherical potential [42]; ●, using A_{12}^* deduced from the kinetic theory of polyatomic gases for the nonspherical potential [9]

the spherical potentials [42] or calculated by Crusius *et al.* [8] and by Hellmann [9] for the nonspherical potentials of the interaction between the molecules by means of the kinetic theory for polyatomic gases, is examined for the interaction viscosity $\eta_{12}^{(0)}$ and for the product $(\rho D_{12})^{(0)}$ in Figs. 3 and 5 as well as in Figs. 4 and 6, respectively. Furthermore, the argumentation of the authors of Ref. [43] that the determination of self-diffusion and binary diffusion coefficients by means of their experimental viscosities in the limit of zero density, $\eta_i^{(0)}$ and $\eta_{12}^{(0)}$, respectively, requires the multiplication by sensible A_i^* and A_{12}^* values based on nonspherical potentials for the interaction between the molecules is verified.

5 Summary and Conclusions

Humberg *et al.* [1, 2] reported on viscosity measurements on the two gaseous mixtures carbon dioxide–nitrogen and ethane–methane including the pure gases applying a rotating-cylinder viscometer between 0.1 MPa and 2.0 MPa in the temperature range of 253.15 K to 473.15 K. The isothermal viscosity data were evaluated by these authors with density series restricted to at most the second order to derive the zero-density viscosities and initial density viscosity coefficients. In the present paper, these data were used to determine the zero-density limits of the interaction

viscosity, $\eta_{12}^{(0)}$, and the product of molar density and diffusion coefficient, $(\rho D_{12})^{(0)}$. Because the data of Humberg *et al.* for a specific mole fraction were not determined at exactly the same temperatures, in a first step, quasi-isotherms were created for the three mole fractions of the mixtures and for the pure gases.

Two procedures were used to deduce $\eta_{12}^{(0)}$. In the first procedure, $\eta_{12}^{(0)}$ was obtained from the values of the zero-density viscosities of the mixtures $\eta_{\text{mix}}^{(0)}(T)$ and the respective values of the pure gases, $\eta_1^{(0)}$ and $\eta_2^{(0)}$, using the Chapman–Enskog theory of dilute monatomic gases [19] in its first-order approximation. The quantity A_{12}^* appearing in this relationship represents the ratio of two effective cross-sections for viscosity and diffusion and can be computed for the effective spherical potential of the interaction of the molecules or, if one applies the more appropriate kinetic theory of polyatomic gases [24, 25] instead of the Chapman–Enskog theory, for a nonspherical potential [8, 9]. However, it was found that $\eta_{12}^{(0)}$ is fairly insensitive to A_{12}^* , so that the differences between $\eta_{12}^{(0)}$ values obtained with A_{12}^* values for the spherical and nonspherical potentials are almost negligible. The obtained $\eta_{12}^{(0)}$ values agree very well with those calculated theoretically, despite the fact that $\eta_{12}^{(0)}$ and A_{12}^* are only defined in the first-order kinetic theory approximation and the experimentally based $\eta_{12}^{(0)}$ values thus represent effective quantities that implicitly account for higher-order kinetic theory contributions. The experimentally based $\eta_{12}^{(0)}$ values are better represented by the theoretically calculated ones for the mixture ethane–methane (within 0.3 %) than for the mixture carbon dioxide–nitrogen (within 0.9 %), with the relative differences being negative for the former and positive for the latter.

The second procedure to determine $\eta_{12}^{(0)}$ uses a modification of the Enskog theory of hard spheres based on the initial density viscosity coefficients for the mixtures and the pure gases. In Ref. [15], the uncertainties of the initial density quantities were found to be too large for obtaining reasonable results for the interaction viscosity, and, unfortunately, in the present paper this problem was encountered as well, even though A_{12}^* values for the nonspherical potentials were used. The $\eta_{12}^{(0)}$ values are higher by up to 40 % for the carbon dioxide–nitrogen system and by up to 20 % for the ethane–methane system than the respective $\eta_{12}^{(0)}$ values obtained with the first procedure. It is therefore concluded that the approximations included in the Enskog theory for binary hard-spheres mixtures at moderate densities are unsuitable to represent the viscosity of a real binary gas mixture.

The experimentally based data of the product $(\rho D_{12,\text{exp}})^{(0)}$, applying the $\eta_{12}^{(0)}$ values from the first procedure and A_{12}^* values derived from the corresponding states principle using universal correlations for a spherical potential [42] as well as A_{12}^* values computed for the nonspherical potential, are compared with theoretical values $(\rho D_{12,\text{cal}})^{(0)}$ computed for the nonspherical potential by means of the kinetic theory for polyatomic gases in its third-order approximation [8, 9]. While the relative deviations of $(\rho D_{12,\text{exp}})^{(0)}$ from $(\rho D_{12,\text{cal}})^{(0)}$ for the spherical potentials reach up to -5.4% for the system carbon dioxide–nitrogen and up to -3.7% for the system ethane–methane, the deviations for the nonspherical potentials are considerably reduced, being within 0.33% to 0.53% for the system carbon dioxide–nitrogen and within -0.05% to -0.30% for the system

ethane–methane. Therefore, the traditional way of estimating self-diffusion and binary diffusion coefficients from experimental viscosity data in the limit of zero density should use suitable values based on nonspherical potentials for A_i^* and A_{12}^* , respectively, whenever possible.

Acknowledgements We would like to thank Prof. Markus Richter from the Chemnitz University of Technology for his approval to use their viscosity data as the basis for the present investigation.

Author Contributions EV wrote the main manuscript and prepared the figures. EB performed the calculations regarding the interaction viscosities and the diffusion coefficients. RH carried out the theoretical calculations for the nonspherical potentials. All authors reviewed the manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL.

Declarations

Competing interests The authors declare no competing interests.

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References

1. K. Humberg, M. Richter, J.P.M. Trusler, R. Span, J. Chem. Thermodyn. **120**, 191 (2018)
2. K. Humberg, M. Richter, J.P.M. Trusler, R. Span, J. Chem. Thermodyn. **147**, 106104 (2020)
3. M. Schäfer, M. Richter, R. Span, J. Chem. Thermodyn. **89**, 7 (2015)
4. R. Hellmann, Mol. Phys. **111**, 387 (2013)
5. R. Hellmann, Chem. Phys. Lett. **613**, 133 (2014)
6. R. Hellmann, E. Bich, E. Vogel, A.S. Dickinson, V. Vesovic, J. Chem. Phys. **129**, 064302 (2008)
7. R. Hellmann, J. Chem. Eng. Data **63**, 470 (2018)
8. J.P. Crusius, R. Hellmann, J.C. Castro-Palacio, V. Vesovic, J. Chem. Phys. **148**, 214306 (2018)
9. R. Hellmann, J. Chem. Thermodyn. **134**, 175 (2019)
10. A. Laesecke, C.D. Muzny, J. Phys. Chem. Ref. Data **46**, 013107 (2017)
11. A. Laesecke, C.D. Muzny, Int. J. Thermophys. **38**, 182 (2017)
12. A. Laesecke, C.D. Muzny, Int. J. Thermophys. **39**, 52 (2018)
13. R. Hellmann, J. Chem. Eng. Data **65**, 968 (2020)
14. Y. Abe, J. Kestin, H.E. Khalifa, W.A. Wakeham, Ber. Bunsenges. Phys. Chem. **54**, 1 (1979)
15. E. Vogel, Int. J. Thermophys. **44**, 75 (2023)
16. S. Hendl, E. Vogel, High Temp. High Press. **25**, 279 (1993)
17. C. Küchenmeister, E. Vogel, J. Baranski, High Temp. High Press. **33**, 659 (2001)
18. V. Teske, E. Vogel, Fluid Phase Equilib. **303**, 126 (2011)
19. G.C. Maitland, M. Rigby, E.B. Smith, W.A. Wakeham, *Intermolecular Forces: Their Origin and Determination* (Clarendon Press, Oxford, 1987)
20. W. Cencek, M. Przybytek, J. Komasa, J.B. Mehl, B. Jeziorski, K. Szalewicz, J. Chem. Phys. **136**, 224303 (2012)
21. E. Bich, R. Hellmann, E. Vogel, Mol. Phys. **106**, 813 (2008)
22. E. Vogel, B. Jäger, R. Hellmann, E. Bich, Mol. Phys. **108**, 3335 (2010)

23. R.F. Berg, M.R. Moldover, J. Phys. Chem. Ref. Data **41**, 043104 (2012)
24. F.R.W. McCourt, J.J.M. Beenakker, W.E. Köhler, I. Kuščer, *Nonequilibrium Phenomena in Polyatomic Gases, Vol. I: Dilute Gases* (Clarendon Press, Oxford, 1990)
25. A.S. Dickinson, R. Hellmann, E. Bich, E. Vogel, Phys. Chem. Chem. Phys. **9**, 2836 (2007)
26. R. Hellmann, E. Bich, E. Vogel, V. Vesovic, J. Chem. Phys. **141**, 224301 (2014)
27. R. Hellmann, E. Bich, V. Vesovic, J. Chem. Thermodyn. **102**, 429 (2016)
28. R. Hellmann, J. Chem. Eng. Data **63**, 246 (2018)
29. R. Hellmann, Fluid Phase Equilib. **485**, 251 (2019)
30. R. Hellmann, Z. Phys. Chem. **233**, 473 (2019)
31. E.F. May, M.R. Moldover, R.F. Berg, Int. J. Thermophys. **28**, 1085 (2007)
32. D.G. Friend, J.C. Rainwater, Chem. Phys. Lett. **107**, 590 (1984)
33. J.C. Rainwater, D.G. Friend, Phys. Rev. A **36**, 4062 (1987)
34. E. Bich, E. Vogel, in *Transport Properties of Fluids. Their Correlation, Prediction, and Estimation, Chap. 5.2–Dense Fluids. Initial Density Dependence*, ed. by J. Millat, J.H. Dymond, C.A. Nieto de Castro (Cambridge University Press, Cambridge, 1996), pp. 72–82
35. R. Di Pippo, J.R. Dorfman, J. Kestin, H.E. Khalifa, Physica A **86**, 205 (1977)
36. J. Kestin, O. Korfali, J.V. Sengers, R. Kamgar-Parsi, Physica A **106**, 415 (1981)
37. V. Vesovic, W.A. Wakeham, Int. J. Thermophys. **10**, 125 (1989)
38. V. Vesovic, W.A. Wakeham, Chem. Eng. Sci. **44**, 2181 (1989)
39. E. Vogel, K. Dobbert, K. Meissner, U. Ruh, E. Bich, Int. J. Thermophys. **12**, 469 (1991)
40. S. Hendl, E. Vogel, Int. J. Thermophys. **16**, 1245 (1995)
41. W.A. Wakeham, V. Vesovic, E. Vogel, S. Hendl, in *Transport Properties of Fluids. Their Correlation, Prediction, and Estimation, Chap. 15–Binary Mixtures: Carbon Dioxide–Ethane*, ed. by J. Millat, J.H. Dymond, C.A. Nieto de Castro (Cambridge University Press, Cambridge, 1996), pp. 388–399
42. E. Bich, J. Millat, E. Vogel, Wiss. Z. W. Pieck Univ. Rostock Naturwiss. Reihe **36**, 5 (1987)
43. E. Bich, J.B. Mehl, R. Hellmann, V. Vesovic, in *Experimental Thermodynamics Volume IX. Advances in Transport Properties of Fluids, Chap. 7–Dilute Gases*, ed. by M.J. Assael, A.R.H. Goodwin, V. Vesovic, W.A. Wakeham (The Royal Society of Chemistry, Cambridge, 2014), pp. 226–252
44. J.H. Dymond, K.N. Marsh, R.C. Wilhoit, K.C. Wong, Virial Coefficients of Pure Gases and Mixtures, in *Landolt-Börnstein: Numerical Data and Functional Relationships in Science and Technology, New Series, Group IV: Physical Chemistry*, vol. 21. (Springer-Verlag, Berlin, 2002)
45. H.L. Robjohns, P.J. Dunlop, Ber. Bunsenges. Phys. Chem. **88**, 1239 (1984)
46. P.S. Arora, H.L. Robjohns, T.N. Bell, P.J. Dunlop, Aust. J. Chem. **33**, 1993 (1980)

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